# How-To: The Plugin System

ragent can load and run third-party **plugins written for OpenAI Codex and Claude Code / Claude Desktop**. Plugin JavaScript executes on an embedded, budget-sandboxed engine (`rquickjs`) built into the ragent binary - **no Node.js, Deno, or Bun installation is required**. Enabled plugins contribute tools (registered as `plugin_<id>_<tool>`) and slash commands through a versioned `ragent` host API.

This document is the full manual: what the system is, how to install and manage plugins, how to write your own, and how to acquire plugins from the existing Claude and Codex ecosystems.

## Table of Contents


- 1. Overview
- 2. Architecture
- 3. The Plugin Store
- 4. Plugin Dialects and Manifests
- 5. The `ragent` Host API (version 1)
- 6. Permissions and the Sandbox
- 7. Managing Plugins
- 8. Writing Your Own Plugin
- 9. Acquiring Plugins from the Claude and Codex Stores
- 10. Configuration Reference
- 11. Security Model
- 12. Error Handling and Lifecycle States
- 13. Performance Budgets
- 14. Troubleshooting
- 15. Reference


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## 1. Overview

### Why it exists

Codex and Claude Code/Desktop have growing third-party plugin ecosystems. Users who have already installed or authored plugins for those tools gain nothing from them inside ragent unless there is a compatibility layer. The plugin system provides that layer:

- it **reuses the existing ecosystem** instead of forcing authors to rewrite plugins ragent-natively;
- it lets **ragent-native plugins** be written once in JavaScript against a documented host API;
- it keeps the ragent single-binary, zero-runtime-dependency promise by running JavaScript in-process on `rquickjs`.

### What a plugin can do

A loaded plugin contributes:

| Contribution | Registered as | Visible in |
|---|---|---|
| **Tools** | `plugin_<id>_<tool>` | `/tools`, the model's tool list, the normal permission system |

| Contribution | Registered as | Visible in |
|---|---|---|
| **Slash commands** | the declared trigger (e.g. `/todo-add`) | The `/` autocomplete menu, the command dispatcher |

and can call a narrow, versioned host API to read plugin-scoped config, emit messages, write log lines, and read files **inside its own directory only**.

#### What a plugin cannot do

- No filesystem access outside the plugin's own directory.
- No network access, no environment variables, no process spawning.
- No `node_modules`, no npm resolution, no CommonJS `require` of arbitrary packages.
- No access to ragent credentials, memory database handles, or session internals.
- No execution at all until the plugin is explicitly **enabled** (discovery and listing parse manifests only).

#### Key facts

- Host API is **version 1**, additive-only.
- A plugin is **enabled when installed** (`/plugins add` records the enable flag), so it loads and contributes at the next session start; `/plugins disable` turns it off. An existing plugin whose ledger row is absent stays disabled.
- **Disable takes effect immediately** in the running session (no restart).
- Plugin code is untrusted by default; every failure is contained so it can never terminate the ragent process.

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## 2. Architecture

The system lives in the **`ragent-plugins`** crate. It deliberately does **not** depend on `ragent-agent` (the tool registry) or `ragent-tui` (the command surface); the session passes closures instead, so the crate stays a bounded, testable unit.

| Module | Responsibility |
|---|---|
| `descriptor` | `PluginDescriptor` model + dialect recognition |
| `manifest` | per-dialect parsing, normalisation, host-API version check |
| `store` | store paths, discovery scan, state ledger (`_state.json`) |
| `add` | `/plugins add` source handling (directory, zip/tar.gz, https URL) |
| `remove` | `/plugins remove` store operation |
| `report` | `/plugins add` and `/plugins remove` report rendering |
| `commands` | `/plugins add\|remove` parse + dispatch glue |
| `control` | `/plugins list\|enable\|disable` |

| Module | Responsibility |
|---|---|
| `runtime` | sandboxed JavaScript runtime (`RuntimePool`, `SandboxContext`) |
| `host_api` | versioned `ragent` host-API bridge |
| `lifecycle` | enable/disable/load/unload/errored lifecycle |
| `tool_adapter` | plugin tools in the session registry |
| `command_adapter` | plugin slash commands in the command surface |
| `session` | session start: discover, load, register, shutdown |
| `harness` | `/plugins test` isolated harness |
| `help` | usage text + subcommand metadata |
| `error` | contained error reporting |

**Runtime design**

- `RuntimePool::checkout(budget)` returns a **fresh** `SandboxContext` pre-configured with the configured memory ceiling and an interrupt handler armed against a caller-supplied deadline. Runtimes are allocated lazily per checkout so a crashed plugin cannot taint a pooled runtime.
- Deadline interruption maps to `PluginError::Timeout`; allocation-failure aborts map to `PluginError::MemoryLimit`; uncaught JavaScript exceptions map to `PluginError::Script`. No panic path escapes.
- The context exposes **no globals beyond what `host_api` installs**: the QuickJS `std`/`os` modules are not registered and `print` is absent.

**Why `rquickjs`**

Both candidate engines (Boa and `rquickjs`) were evaluated against three criteria:

| Criterion | rquickjs | Boa |
|---|---|---|
| Deadline interruption | `Runtime::set_interrupt_handler` aborts `while(true){}` within ~50 ms of the deadline | No wall-clock deadline hook |
| Memory ceiling | `Runtime::set_memory_limit(...)` contains unbounded allocation loops | No allocation-budget API |
| JSON marshalling | JSON crosses as strings via `JSON.parse`/`JSON.stringify` | Equivalent |

`rquickjs` (QuickJS) was selected because the quality of its deadline and memory-ceiling hooks is what the sandbox depends on.

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## 3. The Plugin Store

Plugins live in a **store**. There are two store roots, scanned in ascending-priority order (project wins on plugin-id collision):

1. **user-global**: `~/.config/ragent/plugins/`

2. **project-local**: `<working dir>/.ragent/plugins/`

`plugins.store_dir` in `ragent.json`, when set, **replaces** the project leg.

### Layout

Each plugin is its own subdirectory named after its plugin id:

```
.ragent/plugins/
|-- _state.json            <- per-store enable/disable state + telemetry ledger
|-- codex-weather/
|   |-- codex-plugin.json
|   `-- main.js
`-- claude-todo/
    |-- .claude-plugin/
    |   `-- plugin.json
    `-- index.js
```

### The state ledger (`_state.json`)

Enable/disable state and telemetry counters persist in a per-store `_state.json` file keyed by plugin id, so state survives restarts:

```json
{
  "plugins": {
    "codex-weather": {
      "enabled": true,
      "counters": {
        "loads_ok": 3,
        "load_failures": 0,
        "tool_invocations": 12,
        "tool_failures": 0,
        "consecutive_failures": 0
      }
    }
  }
}
```

Ids absent from the ledger are treated as **disabled with zero counters** (the default); `/plugins add` records a freshly installed plugin **enabled**, so it loads at the next session start until explicitly disabled.

### Discovery rules

- Discovery **parses manifests only**; it never executes JavaScript.
- Unrecognised entries (plain directories, the `_state.json` ledger, dotfiles) are skipped.
- Symlinked plugin directories are followed **only when** their canonical target stays inside the store directory they were found in.
- Scanning never fails wholesale: an absent or unreadable store is treated as empty, and individual bad manifests are reported per-row instead of failing the scan.

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## 4. Plugin Dialects and Manifests

A directory carries **exactly one** dialect. A directory whose contents match both dialects is ambiguous and rejected — **except** the multi-target case: a directory carrying *both nested* manifests (`.codex-plugin/plugin.json` **and** `.claude-plugin/plugin.json`) is a single upstream tree shipping one manifest per host, so ragent resolves it deterministically to the **Claude** dialect (the richer of the two). Any other both-match — a top-level `codex-plugin.json` beside a `claude-plugin.json`, or a `plugin.json` with a `"codex"` marker beside a Claude manifest — stays ambiguous.

### 4.1 Dialect recognition

| Dialect | Recognised by |
|---|---|
| **Codex** | a `codex-plugin.json`, a `.codex-plugin/plugin.json`, **or** a `plugin.json` carrying a top-level `"codex"` marker field |
| **Claude** | a `claude-plugin.json`, **or** a `.claude-plugin/plugin.json` |

### 4.2 Codex manifest (`codex-plugin.json`)

```
{
  "id": "codex-weather",
  "name": "Codex Weather",
  "version": "1.2.0",
  "entry": "main.js",
  "api_version": 1,
  "tools": [
    {
      "name": "get_weather",
      "description": "Return canned weather conditions for a city",
      "parameters": {
        "type": "object",
        "properties": {
          "city": { "type": "string", "description": "City name to look up" }
        },
        "required": ["city"]
      }
    }
  ],
  "commands": [
    { "name": "weather", "description": "Show weather", "usage": "/weather <city>" }
  ]
}
```

| Field | Type | Required | Meaning |
|---|---|---|---|
| `name` | string | yes | Display name |

| Field | Type | Required | Meaning |
|---|---|---|---|
| `version` | string | yes | Version string |
| `entry` | string | yes | Entry-point JS file, relative to the plugin root |
| `id` | string | no | Explicit plugin id; derived from `name` when absent |
| `api_version` | u32 | no | Declared host-API version; defaults to 1 |
| `tools[]` | array | no | Tools declared in the manifest |
| `commands[]` | array | no | Slash commands declared in the manifest |
| `permissions.network[]` | array | no | Network grants (e.g. `"outbound"` becomes `network.outbound`) |
| `permissions.fs` | object | no | Filesystem grants - **recorded as unsupported** |
| `permissions.exec` | object | no | Subprocess grants - **recorded as unsupported** |
| `mcp_servers` | object | no | MCP transports - **recorded as unsupported** |

### 4.3 Claude manifest (`.claude-plugin/plugin.json`)

```
{
  "name": "Claude Todo",
  "version": "0.3.0",
  "entry": "index.js",
  "api_version": 1,
  "tools": [
    {
      "name": "add_todo",
      "description": "Add an item to the plugin's todo list",
      "parameters": {
        "type": "object",
        "properties": { "text": { "type": "string" } },
        "required": ["text"]
      }
    }
  ],
  "commands": [
    { "name": "todo-add", "description": "Add a todo item", "usage": "/todo-add <text>" }
```

```
  ]
}
```

| Field | Type | Required | Meaning |
|---|---|---|---|
| `name` | string | yes | Display name |
| `version` | string | no | Defaults to `0.0.0` |
| `entry` | string | conditional | Entry point |
| `main` | string | conditional | Entry-point fallback |
| `server.entry` | string | conditional | Claude Desktop entry-point fallback |
| `id` | string | no | Explicit id; derived from `name` when absent |
| `api_version` | u32 | no | Declared host-API version; defaults to 1 |
| `tools[]` | array | no | Manifest-declared tools |
| `commands[]` | array | no | Manifest-declared commands |
| `permissions[]` | array | no | Requested permission strings |
| `capabilities[]` | array | no | Entries outside the v1 set are **recorded as unsupported** |
| `mcp_servers` | object | no | MCP transports - **recorded as unsupported** |
| `mounts` | object | no | Claude Desktop mount section - **unsupported** |
| `window` | object | no | Claude Desktop window section - **unsupported** |

**Entry resolution order for Claude:** `entry`, then `main`, then `server.entry`. A manifest with none of these is rejected.

### 4.4 Unsupported capabilities

Manifest sections with no ragent host-API equivalent are **recorded and reported, never silently dropped**. The stable labels surfaced by `/plugins list` and `/plugins test` are:

| Label | Source |
|---|---|
| `mcp server transports` | `mcp_servers` whose entry shape the bridge cannot satisfy (e.g. an array of transports); an inline object or a `"mcpServers": "./mcp.json"` file reference is bridged and connected instead |
| `claude desktop mounts` | Claude `mounts` |
| `claude desktop window` | Claude `window` |

| Label | Source |
|---|---|
| `codex permissions.fs` | Codex `permissions.fs` |
| `codex permissions.exec` | Codex `permissions.exec` |

### 4.5 Tool and command declarations

```
// Tool declaration (manifest or register_tool)
{
  "name": "get_weather",           // unique within the plugin
  "description": "...",            // shown to the model
  "parameters": { ... }            // JSON schema for the arguments object
}

// Command declaration (manifest or register_command)
{
  "name": "todo-add",              // the slash trigger, without the leading '/'
  "description": "...",            // shown in autocomplete
  "usage": "/todo-add <text>"      // optional help hint
}
```

Registered tool names are namespaced **`plugin_<pluginid>_<toolname>`**, so a plugin tool never collides with a built-in by construction; a *duplicate declaration within one plugin* or a *command that collides with a built-in slash trigger* is a genuine collision and is rejected.

### 4.6 Plugin id derivation

When `id` is absent, it is derived from `name`: split on every non-alphanumeric character, drop empties, join with `-`, lowercase. `"Codex Weather"` becomes `codex-weather`; an empty result becomes `unnamed-plugin`.

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## 5. The `ragent` Host API (version 1)

Plugin code accesses exactly these capabilities through the global `ragent` object:

```
ragent.api_version                          -> 1
ragent.plugin_id                            -> string
ragent.register_tool(def)                   -> { name, description, parameters, handler }
ragent.register_command(def)                -> { name, description, usage, handler }
ragent.config.get(key)                      -> JSON value from plugin-scoped config, or undefined
ragent.message.info(text) / .warn(text) / .error(text)
ragent.log(level, text)
ragent.plugin.read_text_file(rel_path)      -> string (restricted to the plugin directory)
```

### 5.1 `register_tool(def)`

Registers a tool whose handler executes when the agent invokes it.

```
ragent.register_tool({
  name: "add_todo",
  description: "Add an item to the plugin's todo list",
  parameters: {
    type: "object",
    properties: { text: { type: "string", description: "Todo item text" } },
    required: ["text"]
  },
  handler: function (args) {
    var text = args && args.text ? String(args.text) : "";
    return { added: true, text: text };
  }
});
```

You may also declare the tool in the manifest and define a **global function named after the tool**; dispatch resolves the handler from `globalThis.__ragent_tools[name]` and falls back to a global function of the same name:

```
// The tool "get_weather" is declared in codex-plugin.json.
function get_weather(args) {
  return { city: args.city, conditions: "clear skies", temperature_c: 17 };
}
```

### 5.2 `register_command(def)`

```
ragent.register_command({
  name: "todo-add",
  description: "Add a todo item",
  usage: "/todo-add <text>",
  handler: function (text) {
    return "Added todo: " + text;
  }
});
```

Argument text typed after the trigger is passed to the handler as a plain JavaScript string; the handler's return value is surfaced as message-window text (bare strings pass through, `undefined`/`null` produce empty text, anything else is `JSON.stringify`-compacted).

### 5.3 Marshalling rules

Tool arguments cross into the sandbox as a JSON document bound to the scratch global `__ragent_plugin_args` and handed to the handler as a plain JavaScript value. The handler's return value is interpreted as:

| Return value | Result |
|---|---|
| `undefined` / `null` | empty content |
| a bare string | passed through as content |
| `{ content, isError? }` | unwrapped into content plus an error flag (`isError: true` fails the call) |
| any other value | surfaced compactly as content and carried under tool metadata |

If `JSON.stringify` of the return value throws (for example a cyclic object), the tool call fails with a serialisation error rather than panicking.

### 5.4 Version rule

A plugin whose declared `api_version` is **newer** than the host's is refused at enable/test time with a version-mismatch report. Equal or lower versions are accepted (v1 is additive-only, so older declarations keep working).

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## 6. Permissions and the Sandbox

### 6.1 Capability groups

The host API is installed as capability groups, each keyed by one of:

`tools, commands, config, message, log, plugin.read_text_file`

Before installing a group, the runtime asks a **permission gate** for a decision. An ungranted group is **simply absent** from the injected `ragent` object. The full permission key for a capability is:

`plugin:<pluginid>:<capability>`

### 6.2 How gates are built

The gate is built from the plugin's declared permissions plus the `plugins.permissions` config grants. In `ragent.json`:

```
{
  "plugins": {
    "permissions": {
      "codex-weather": ["tools", "commands", "message", "log"],
      "experimental": ["*"]           // "*" grants every capability group
    }
  }
}
```

- A plugin with no matching entry gets the **full v1 surface** currently (no gate is installed). Grant an explicit list to restrict a specific plugin.
- `"*"` grants every capability group.

### 6.3 Sandbox budgets

| Budget | Config key | Default | Meaning |
|---|---|---|---|
| Per-tool wall clock | `max_execution_ms` | 5000 ms | Maximum time a single tool/command handler may run |
| Entry-point wall clock | `max_entry_ms` | 10000 ms | Maximum time the entry point may run |
| Memory ceiling | `max_memory_mb` | 64 MiB | Per-JavaScript-context heap ceiling |

Deep recursion trips a 1 MiB engine stack before the heap ceiling, so plugin code cannot overflow the host stack via JS recursion.

### 6.4 What the sandbox denies

- Filesystem access outside the plugin directory (`plugin.read_text_file("../../etc/passwd")` is refused; `..` components and absolute paths are rejected before any read).
- Network access of any kind.
- Environment variables and process spawning.
- Everything not explicitly exposed through the `ragent` object.

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## 7. Managing Plugins

The same operations are available on two surfaces that share one help text and one set of handlers:

- the **TUI slash command**: `/plugins <sub> [args...]`
- the **CLI parity surface**: `ragent plugins <sub> [args...]`

### 7.1 Subcommands

```
/plugins list [--verbose]        # list discovered plugins
/plugins add <source> [--force]  # install a plugin (enabled on install)
/plugins remove <pluginid>       # uninstall (refused while enabled)
/plugins enable <pluginid>       # enable, load, register tools/commands
/plugins disable <pluginid>      # unload, deregister tools/commands
/plugins test <pluginid>         # isolated harness: load + invoke each tool once
/plugins help                    # usage block
```

| Form | Description |
|---|---|
| `list [--verbose]` | One row per discovered plugin: id, name, version, dialect, state (`disabled`/`enabled`/`loaded`/`errored`), tool and command counts, plus a totals summary. `--verbose` (alias `-v`) appends per-plugin telemetry counters. |
| `add <source> [--force]` | Install and validate a plugin; reports id and dialect. Recorded **enabled**, so it loads at the next session start. Refuses a duplicate id unless `--force`. |
| `remove <pluginid>` | Uninstall from the store. **Refused while the plugin is enabled.** |
| `enable <pluginid>` | Mark enabled, load into the current session, register tools and commands. States the declared permissions and the outcome. |
| `disable <pluginid>` | Mark disabled, unload, deregister every contributed tool and command, confirming how many of each were removed. Files are **not** deleted. |

| Form | Description |
|---|---|
| `test <pluginid>` | Load in an isolated harness (never touching the live session), invoke each tool once with schema-derived sample arguments, and report per-step results. |
| `help` | Print the usage block. A bare `/plugins` or an unknown subcommand does the same, creating no files. |

### 7.2 Sources accepted by `add`

- a local **directory** containing a plugin manifest;
- a local `.zip` or `.tar.gz` package file;
- an **`https://` URL** pointing at a `.zip`/`.tar.gz` package (non-`https` URLs are refused);
- a **`git+<https-url>#<ref>[:<subpath>]`** git source - the form the store browse panels emit (see section 9). A `:<subpath>` source is a shallow **sparse** clone of that subdirectory, so a plugin in a monorepo installs without downloading the whole tree; a whole-repo source (no subpath, the shape a Claude marketplace `{"source":"./"}` resolves to for a plugin published at its repository root) is a shallow clone of the **complete** tree, so a manifest in the nested `.claude-plugin/` directory is found.

A single-root archive wrapper (e.g. GitHub's `<repo>-<ref>/`) is descended automatically. Archives are capped at **50 MiB** and extracted with path-traversal protection (no `..` or absolute entries).

### 7.3 Worked examples (TUI)

```
/plugins help
/plugins list
/plugins add ./vendor-plugins/codex-weather
/plugins add ./packages/claude-todo.zip
/plugins add https://example.com/plugins/plugin.tar.gz
/plugins list --verbose
/plugins enable codex-weather
/plugins test claude-todo
/plugins disable codex-weather
/plugins remove claude-todo
```

### 7.4 CLI parity

```
ragent plugins list
ragent plugins add ./my-plugin
ragent plugins add https://example.com/pkg.zip --force
ragent plugins enable my-plugin
ragent plugins test my-plugin
ragent plugins disable my-plugin
ragent plugins remove my-plugin
ragent plugins help
```

The CLI rewrites the TUI attribution lines: `From: /plugins ...` becomes a plain `ragent plugins ...` header and `## /plugins` becomes `## ragent plugins`.

### 7.5 Output shape

Reports are prefixed with a `From: /plugins <sub>` attribution line. `list` renders a fixed-width table with a count column per contribution kind (tools, commands, skills, agents, hooks):

```
| ID                           | Name                 | Version  | Dialect | State    | Tools | C
|------------------------------|----------------------|----------|---------|----------|-------|--
| codex-weather                | Codex Weather        | 1.2.0    | codex   | loaded   |     1 |
| claude-tools                 | Claude Tools         | 2.0.0    | claude  | loaded   |     0 |
```

followed by optional `Contributions:`, `Unsupported capabilities:`, and `Errors:` sections, telemetry lines in verbose mode, and a summary:

```
Contributions:
- claude-tools: tools []; commands []; skills [db-setup]; agents [agents/security-reviewer.md]; h

Total: 2 plugin(s) - 2 enabled, 0 disabled, 0 errored.
```

`test` renders one line per harness step:

```
[ ok ] discovery (2 ms)
[ ok ] manifest validation (0 ms)
[ ok ] version check (0 ms)
[ ok ] entry execution (5 ms)
[ ok ] sample invocation add_todo (1 ms)
```

and ends by confirming the harness was unloaded with the live session untouched.

---

## 8. Writing Your Own Plugin

### 8.1 Minimal Codex plugin

Create a directory with two files.

**`codex-plugin.json`**

```json
{
  "id": "hello-codex",
  "name": "Hello Codex",
  "version": "1.0.0",
  "entry": "main.js",
  "api_version": 1,
  "tools": [
    {
      "name": "greet",
      "description": "Greet a person by name",
      "parameters": {
        "type": "object",
        "properties": { "who": { "type": "string" } },
        "required": ["who"]
      }
    }
```

```
  ]
}
```

**main.js**

```
// Entry-point top-level code runs once, when the plugin is enabled.
ragent.message.info("hello-codex loaded");

// The tool "greet" is declared in the manifest; dispatch finds this global.
function greet(args) {
  var who = args && args.who ? String(args.who) : "world";
  return { greeting: "Hello, " + who + "!" };
}
```

Install and enable it:

```
/plugins add ./hello-codex
/plugins enable hello-codex
```

The model can now call `plugin_hello-codex_greet`.

### 8.2 Minimal Claude plugin

**.claude-plugin/plugin.json**

```
{
  "name": "Hello Claude",
  "version": "1.0.0",
  "entry": "index.js",
  "api_version": 1
}
```

**index.js**

```
ragent.register_tool({
  name: "shout",
  description: "Uppercase a message",
  parameters: {
    type: "object",
    properties: { text: { type: "string" } },
    required: ["text"]
  },
  handler: function (args) {
    return String(args.text || "").toUpperCase();
  }
});

ragent.register_command({
  name: "hello",
  description: "Say hello",
  usage: "/hello <name>",
  handler: function (who) {
    return "Hello, " + (who || "world") + "!";
```

```
  }
});
```

### 8.3 Reading plugin-scoped config

Grant the `config` capability, then read keys the host provides:

```
var city = ragent.config.get("default_city");   // undefined when absent
```

### 8.4 Emitting messages and logs

```
ragent.message.info("informational note in the message window");
ragent.message.warn("a warning");
ragent.message.error("an error");
ragent.log("debug", "goes to the host log, not the message window");
```

### 8.5 Reading a file inside the plugin directory

```
// Allowed: a file inside the plugin's own directory.
var data = ragent.plugin.read_text_file("data/seed.json");

// Refused: '..' components and absolute paths are rejected before any read.
// var nope = ragent.plugin.read_text_file("../../outside.txt");
```

### 8.6 Test before enabling

`/plugins test <id>` loads the plugin in an isolated harness, executes the entry point, and invokes each contributed tool once with schema-derived sample arguments. It never touches the live session, never writes a ledger row, and never leaves the plugin enabled. Use it as the first check after any edit.

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## 9. Acquiring Plugins from the Claude and Codex Stores

ragent installs plugins from a directory, a local archive, or an `https://` URL with `/plugins add <source>`, and it also browses each store's official marketplace from the TUI. Two subcommands open an interactive browse panel over a store index:

- `/plugins codex [query] [--category <name>] [--refresh]` - the Codex marketplace.
- `/plugins claude [query] [--category <name>] [--refresh]` - the Claude marketplace.

Each panel reads the store index (normalised by that store's `StoreProvider`), lists the plugins it declares, narrows by the typed `query`, and installs the selected plugin on `Enter`. `--refresh` bypasses the on-disk store cache and `--category <name>` pre-selects a category row once the index has landed.

A vendor marketplace is a *listing*: the Codex catalogue carries neither a version nor a description, and the Claude catalogue carries a description but only sometimes a version. After the index lands, the panel resolves each plugin's `version` and `description` from the plugin's **own manifest**, fetched off the event loop from the raw URL derived from its install source (a `github.com` git source), with the store's dialect-appropriate manifest name tried first. The title shows `loading details...` while that runs, and a version that remains unknown renders as `-`. The lookups are bounded in concurrency and size, cached across opens, and never fatal: a plugin with no derivable manifest (an archive source, a non-GitHub host) keeps the listing's own fields. No plugin JavaScript is executed to read a manifest.

### 9.0 Adding a marketplace

A store does not only browse its one official marketplace: `/plugins marketplace <store> add <source>` registers additional marketplaces, and the two browse panels fetch **all** of them and merge the entries into the one panel (de-duplicated by plugin id). `<store>` is `codex` or `claude`, and `<source>` takes any of the forms the Claude Code `/plugin marketplace add` prompt accepts:

| Source form | Resolves to |
|---|---|
| `owner/repo` | `https://raw.githubusercontent.com/owner/repo/main/<store marketplace document>` (`.agents/plugins/marketplace.json` for Codex, `.claude-plugin/marketplace.json` for Claude). |
| `git@github.com:owner/repo.git` | The same raw document as `owner/repo`. |
| `https://.../marketplace.json` | Used verbatim (any `https` marketplace-index URL). |
| `./path/to/marketplace` | A local marketplace directory; its manifest `name` is recorded and it is not fetched remotely. |

```
/plugins marketplace claude add anthropics/claude-plugins-community
/plugins marketplace claude add https://example.org/internal/marketplace.json
/plugins marketplace claude add ./my-marketplace
/plugins marketplace claude list                  # base endpoint + added marketplaces
/plugins claude                                   # now lists both marketplaces' plugins
/plugins marketplace claude remove claude-plugins-community
```

The marketplace is written to the **user-global** config (`plugins.stores.marketplaces` in `~/.config/ragent/ragent.json`), never the untrusted project config, and is picked up by the next `/plugins <store>` browse. `--force` replaces an existing marketplace with the same name; without it a duplicate name is refused. `/plugins marketplace show` lists every configured marketplace for **both** stores in one section each; `/plugins marketplace <store> list` lists one store's. Both are pure config reads (no network). A base endpoint that cannot be reached renders the inline "unavailable" row; an added marketplace URL that is refused (non-`https`) is skipped rather than blinding the whole store.

### 9.1 The category navigator

Both panels carry a left-hand **category navigator**: `ALL` first, then the distinct non-empty categories the fetched index declares (sorted case-insensitively). An entry's `tags` are query-search terms only and never form a navigator row (FR-021 withdrawn), so a plugin carrying a long `keywords` list adds a single category row at most. It is driven by keyboard and mouse:

| Input | Effect |
|---|---|
| `Tab` | Move focus between the navigator and the result list. |
| `Up` / `Down` | Move the focused pane's cursor. Moving the category cursor applies the highlighted category on the same keypress. |

| Input | Effect |
|---|---|
| `Enter` | Over the navigator, apply the category and return focus to the results (never installs); over a result, install the plugin. |
| `c` | Clear the category back to `ALL`. |
| Click | A left-click on a category row selects and applies it; a left-click on a result row moves the result cursor. |
| Wheel | Scrolling over the navigator moves its viewport without changing the active category. |

Selecting a category re-derives the visible set from the full fetched index in memory (**no network request**) and composes with the free-text query, so a result is visible only when it matches both. A category the index no longer declares is never shown as a row, and a selection that vanishes after a refresh falls back to `ALL`. A store marketplace that cannot be reached renders a clean `<Store> plugin marketplace is unavailable (<plain cause>)` row rather than the raw transport error.

Every mouse path is a convenience over the keyboard path, so a terminal without mouse reporting still reaches every category, and no navigator action ever installs, downloads, or runs plugin JavaScript.

Each result row renders four columns: the plugin **id**, its **version**, the **marketplace it came from**, and its **description**. The marketplace column names the added marketplace an entry was fetched from (the `name` it was added under), or the store label (`Codex`/`Claude`) for the store's base endpoint, so a merged multi-marketplace panel shows each row's source at a glance. The panel widens up to a 140-column ceiling (a third wider than before) and the marketplace and description columns share the flexible remainder, so a longer marketplace name is shown when the terminal has room and clipped gracefully (without pushing the fixed columns or the `[installed]` marker off the row) when it does not.

A fixed column-heading row (`Id`, `Version`, `Marketplace`, `Description`) sits directly above the result list and never scrolls with the rows. It uses the same column template as a result row so each label sits above its column, and is painted dark-on-light (dark text on a light background) so it reads as a pinned label rather than a result row. A click or wheel on the heading row is inert.

The result list is rendered in **ascending alphabetical order of plugin id** (case-insensitively), derived from the id alone rather than the order the index or the marketplace merge delivered the entries in. The order is applied when the index lands, so a free-text query and a category filter both narrow the already-sorted set and the survivors keep their alphabetical order; the navigator's category rows are sorted separately (`ALL` first, then the distinct categories case-insensitively), and a case-insensitive category collision (for example `Utility` and `utility`) keeps the first spelling the index declares.

### 9.2 Claude Code / Claude Desktop plugins

Claude Code plugins are bundled, shareable units that can bundle slash commands, agents, skills, hooks, and MCP servers. They are distributed as **git repositories on GitHub**, and collections of plugins from one source are published as **marketplaces** (also git repositories). Anthropic maintains an official marketplace.

**Step 1 - clone the plugin or marketplace repository.**

```
# A single plugin repository
git clone https://github.com/<owner>/<plugin-repo> vendor/claude-plugin
```

```
# Or the official marketplace, which contains many plugins
git clone https://github.com/anthropics/claude-plugins-official vendor/claude-plugins
```

**Step 2 - inspect the tree for a Claude-dialect manifest.**

A Claude-dialect plugin has either a `.claude-plugin/plugin.json` or a `claude-plugin.json` at its root:

```
ls vendor/claude-plugin/.claude-plugin/plugin.json
# or
ls vendor/claude-plugin/claude-plugin.json
```

If the repository is a marketplace, each plugin is a subdirectory; find the one you want and install that subdirectory.

**Step 3 - install it.**

```
/plugins add ./vendor/claude-plugin
/plugins list
/plugins enable <pluginid>
```

If the plugin declares `mcp_servers`, sections ragent cannot satisfy, or Desktop-only `mounts`/`window` sections, those are reported under `Unsupported capabilities:` in `ragent`'s list output - the rest of the plugin still loads.

### 9.3 OpenAI Codex plugins

Codex plugins are recognised by a `codex-plugin.json`, a `.codex-plugin/plugin.json` (the layout the official `openai/plugins` catalogue ships), or a `plugin.json` with a top-level `"codex"` marker. The Codex plugin packaging format is not yet formally standardised, so ragent's recogniser treats the Codex dialect as the `name`/`version`/`entry` shape documented in section 4.2, with optional `tools[]` and `commands[]`.

**Step 1 - clone the plugin repository** (or download a release archive).

```
git clone https://github.com/<owner>/<codex-plugin-repo> vendor/codex-plugin
```

**Step 2 - confirm the manifest.**

```
ls vendor/codex-plugin/codex-plugin.json    # or .codex-plugin/plugin.json
```

**Step 3 - install it.**

```
/plugins add ./vendor/codex-plugin
/plugins enable <pluginid>
```

### 9.4 Installing from a hosted package (archive or URL)

When the plugin is published as a `.zip` or `.tar.gz`, install it directly - locally or over `https`:

```
# Package a directory you already have
cd vendor && zip -r ../claude-todo.zip claude-todo && cd ..

# Host the archive yourself (any https-served location works)
# e.g. a GitHub release asset URL:

/plugins add https://github.com/<owner>/<repo>/releases/download/v1.0.0/plugin.zip
```

A GitHub source ZIP (`https://github.com/<owner>/<repo>/archive/refs/heads/main.zip`) also works: it is downloaded, extracted, the single wrapper directory (`<repo>-<ref>/`) is descended automatically, and the manifest is validated before anything lands in the store.

### 9.5 Adapting a plugin that is not yet ragent-compatible

Because ragent recognises the **manifest dialects**, many upstream plugins load unchanged. When one does not, the usual fixes are small:

1. **No manifest in a recognised location.** Add `codex-plugin.json`, `.codex-plugin/plugin.json`, or `.claude-plugin/plugin.json` with `name`, `version`, and `entry`.
2. **No JavaScript entry point.** Point `entry` at the plugin's JS file; a hand-written `main.js` that calls `ragent.register_tool` is enough.
3. **A capability ragent does not implement** (MCP transports, subprocess grants, Desktop mounts). Nothing to do - it is reported as unsupported and the plugin still loads for its other contributions.
4. **`api_version` newer than 1.** ragent's host API is version 1; reduce the declaration if the plugin only uses v1 features.

### 9.6 A note on trust

Enabling a third-party plugin is a **trust decision**. Before enabling:

- read the entry-point JavaScript;
- check the declared permissions and grant only what is needed via `plugins.permissions`;
- run `/plugins test <id>` first - it executes the plugin in an isolated harness without registering anything in the live session.

See section 11 for the full security posture.

### 9.7 Capability coverage by store

Codex and Claude plugins ship much the same broad set of contribution surfaces, but each store spells them differently and ragent bridges only some. The table below lists the known surfaces for each store provider (the Codex store `openai/plugins` and the Claude store `claude-plugins-official`, the two documents the store browser parses) and whether ragent consumes them.

"Partial" and "No" rows never block a plugin's other contributions: a surface that has an FR-025 label is recorded and reported under `Unsupported capabilities:`, and any other unrecognised section is simply ignored. The "In ragent?" column answers "if the plugin's manifest is recognised, does ragent consume this surface?".

| Capability | Codex store (`openai/plugins`) | Claude store (`claude-plugins-official`) | In ragent? |
|---|---|---|---|
| **Manifest** | `.codex-plugin/plugin.json` | `.claude-plugin/plugin.json` | Yes - both nested locations are recognised (plus a root `codex-plugin.json` / `claude-plugin.json`, and a `plugin.json` with a top-level `"codex"` marker) |

| Capability | Codex store (`openai/plugins`) | Claude store (`claude-plugins-official`) | In agent? |
|---|---|---|---|
| **JavaScript tools** | - (store entries carry no `entry`) | `entry` / `main` / `server.entry` | **Yes** - the entry point runs on the sandbox; `tools[]` and `register_tool` register as `plugin_<id>_<tool>` |
| **Slash commands** | `commands/` | `commands/`, manifest `commands[]` | **Yes** - prompt commands from `commands/*.md`, plus inline `register_command` commands (FR-031) |
| **Skills** | `skills/` (declared `skills`) | `skills/` (declared `skills`) | **Yes** - each declared skill directory joins skill discovery (FR-029) |
| **MCP servers** | `mcpServers` / `.mcp.json` | `mcpServers` / `.mcp.json` | **Yes** - bridged and connected as `<plugin-id>.<server>` (FR-030) |
| **Agents / subagents** | `agents/` | `agents/` | **Yes** - declared profiles and every `agents/*.md` file join agent discovery, so they appear in the agent picker and are spawnable (FR-032) |

| Capability | Codex store (`openai/plugins`) | Claude store (`claude-plugins-official`) | In ragent? |
|---|---|---|---|
| **Hooks** | `hooks.json` | `hooks/`, `hooks.json` | **Yes** - hooks declared inline or in a `hooks.json` file (plugin root or `hooks/`) are normalised onto the session hook engine (FR-033), including the group `{matcher, hooks: [...]}` shape; the trigger name accepts both ragent's `pre_tool_use` and the Claude `PreToolUse` spellings; a plugin hook gets `CLAUDE_PLUGIN_ROOT` and the Claude event JSON on stdin, and a blocking Stop hook can feed findings back before the turn ends |
| **App / UI surfaces** | `apps`, `.app.json`, `interface` | - | **No** - Codex app/UI metadata is host-specific and ignored |
| **Desktop integration** | - | `mounts`, `window` | **No** - recorded as `claude desktop mounts` / `claude desktop window` |
| **Subprocess / filesystem grants** | `permissions.fs`, `permissions.exec` | - | **No** - recorded as `codex permissions.fs` / `codex permissions.exec` |

| Capability | Codex store (`openai/plugins`) | Claude store (`claude-plugins-official`) | In ragent? |
|---|---|---|---|
| **LSP servers / monitors** | - | LSP servers, monitors | **No** - recorded as `claude lsp servers`; a marketplace `*-lsp` stub whose repo holds no manifest (the inline-`lspServers` shape, e.g. `rust-analyzer-lsp`) installs via a manifest **materialised** from the marketplace document into `.claude-plugin/plugin.json`, so the "no plugin manifest found" refusal now applies only to trees no recognised manifest could be produced for |
| **Non-JS bundle** | common | common | **Yes** - installs and loads inertly, contributing only its bridged skills/MCP (FR-028) |

Two practical consequences:

- **Codex store entries install and load.** The `openai/plugins` catalogue keeps each manifest in `.codex-plugin/plugin.json`, and ragent recognises that nested location just like Claude's `.claude-plugin/plugin.json`, so `/plugins add` and `/plugins enable` both succeed. (A Codex entry whose manifest sits in some other unrecognised location still reports `no plugin manifest found`.)
- **Skills, commands, agents, and hooks are the surfaces that carry most value today**, because a large share of both catalogues are non-JS bundles. A plugin's `commands/*.md` become prompt commands (registered under the bare name when free, otherwise `plugin:<id>:<name>`); its declared `skills` join the session's skill discovery; its `agents/*.md` profiles join agent discovery (YAML frontmatter is accepted); and its declared `hooks` fire on the matching session lifecycle events.

---

## 10. Configuration Reference

The optional `plugins` block in `ragent.json`:

```
{
  "plugins": {
    "enabled": true,                // master switch; default true
```

```
    "max_execution_ms": 5000,        // per-tool wall-clock budget
    "max_entry_ms": 10000,           // entry-point wall-clock budget
    "max_memory_mb": 64,             // per-JS-context memory ceiling
    "store_dir": null,               // optional override of the plugin store path
    "permissions": {                 // optional per-plugin permission grants
      "codex-weather": ["tools", "commands", "message", "log"]
    }
  }
}
```

| Field | Type | Default | Description |
|---|---|---|---|
| `enabled` | bool | `true` | Master switch. When `false`, no discovery or loading happens and `/plugins` subcommands other than `help` report the subsystem is disabled. |
| `max_execution_ms` | u64 | `5000` | Per-plugin-tool wall-clock execution budget, milliseconds. |
| `max_entry_ms` | u64 | `10000` | Entry-point wall-clock budget, milliseconds. |
| `max_memory_mb` | u64 | `64` | Per-JavaScript-context memory ceiling, MiB. |
| `store_dir` | string? | `null` | Override of the plugin store directory. When `null`, the store is `.ragent/plugins/` (project) falling back to `~/.config/ragent/plugins/` (user-global). |
| `permissions` | map<string, string[]> | `{}` | Per-plugin capability grants keyed by plugin id. Use `["*"]` to grant everything. |

The section merges **overlay-wins**: a project-level `plugins` block is not discarded by an absent user-global block. With `plugins.enabled: false`, every subcommand except `help` reports the subsystem is disabled and no plugin code executes.

---

## 11. Security Model

### 11.1 Isolation

- Plugin code runs with **no access** to the ragent process filesystem, network stack, environment variables, or credentials other than what the host API explicitly exposes.

- The host API **never** exposes API keys, provider credentials, memory database handles, or session internals.
- The sandbox is **in-process** (no seccomp, jails, or OS-process isolation).

### 11.2 Installation safety

- Store scanning follows symbolic links only within the store directory itself.
- `/plugins add` from a URL downloads **over `https` only**, to a 50 MiB cap.
- Archives are extracted with **path-traversal protection** (no `..` or absolute entries).

### 11.3 Execution containment

- Discovery and `/plugins list` parse manifests only; nothing executes.
- JavaScript executes only for an **enabled** plugin during session start, or inside `/plugins test`.
- Every plugin failure is contained; a plugin can never terminate the ragent process.
- Host-level traces of capability use and gate decisions are emitted at `debug`/`trace` only - no plugin file contents, tool results, or message payloads are logged above that level.

### 11.4 Trust is explicit

Enabling a plugin is a trust decision. The enable path **states the plugin's declared permissions before completing**, and `/plugins test` lets you exercise a plugin without registering anything in the live session.

---

## 12. Error Handling and Lifecycle States

### 12.1 Lifecycle states

| State | Meaning |
|---|---|
| `disabled` | Installed but inert: no code executes, no contributions (still scans and lists). |
| `enabled` | Marked enabled but not currently loaded in this session. |
| `loaded` | Enabled and loaded; contributes tools and commands. |
| `errored` | A failure prevented loading or registration; contributes nothing. |

### 12.2 Failure handling

| Failure | Effect |
|---|---|
| Manifest parse failure | Plugin marked `errored` with cause `manifest-parse` and the JSON error position; never loads. |

| Failure | Effect |
|---|---|
| Entry-point execution failure | Plugin marked `errored` with cause `entry` and the JS exception message; partial tool registration is deregistered. |
| Tool-invocation failure (exception, timeout, memory ceiling, bad return) | Fails the individual call and is recorded in telemetry; the plugin state is unchanged **unless** consecutive failures reach the auto-unload threshold (default 3), when the plugin is auto-unloaded and marked `errored`. |
| Name collision | Registration is all-or-nothing per plugin; the plugin is marked `errored` with cause `name-collision` and contributes nothing. |
| Version mismatch (`api_version` newer than host) | Refused at enable/test time with a version-mismatch report. |
| `/plugins` argument errors, unknown ids, duplicate add without `--force` | Reported as `[err]` rows in the message window; no state changes. |

### 12.3 Telemetry counters

Recorded per plugin in the state ledger and shown by `list --verbose`:

`loads_ok, load_failures, tool_invocations, tool_failures, consecutive_failures`

---

## 13. Performance Budgets

| Operation | Budget |
|---|---|
| Discovery + manifest parse for up to 50 installed plugins | under 2 s (excluding enabled plugins' entry execution) |
| Enabled plugin entry execution within budget | at most 200 ms amortised to session start per plugin |
| `/plugins test` on a plugin contributing up to 10 tools | under 15 s, excluding user permission prompts |

Observed on a debug build: discovery + list over 50 installed plugins completes in 0.01-0.02 s; `ragent plugins test` on a 10-tool plugin completes in 0.01 s; an infinite-loop entry is contained at the configured entry budget (e.g. 0.8 s at an 800 ms budget).

---

## 14. Troubleshooting

| Symptom | Cause | Fix |
|---|---|---|
| `No plugins discovered.` | Store empty or wrong location | Check `.ragent/plugins/` and `~/.config/ragent/plugins/`; set `plugins.store_dir` if using a custom path |
| Plugin row shows `dialect ?` and version `-` | Manifest failed to parse | Read the `Errors:` section for the JSON position; validate the manifest |
| `ambiguous plugin manifest` | Directory matches both Codex and Claude recognition *and* is not the both-nested-manifest multi-target layout | Remove one of the two manifest forms (a plugin shipping both `.codex-plugin/plugin.json` and `.claude-plugin/plugin.json` is accepted as Claude and needs no change) |
| `plugin add: no plugin manifest found in <dir>` | The installed tree carries no manifest ragent recognises | Check for `.claude-plugin/plugin.json`, `claude-plugin.json`, `.codex-plugin/plugin.json`, `codex-plugin.json`, or a `plugin.json` with a top-level `"codex"` marker. A Claude-dialect plugin's only manifest is the *nested* `.claude-plugin/plugin.json`, so a source that names a subdirectory without it fails: point the source at the directory that actually holds the manifest |
| `plugin requires host API vN` | Declared `api_version` newer than 1 | Lower the declaration to `1` |
| Enable reports cause `entry` | Entry point threw or timed out | Run `/plugins test <id>` for the exception/timeout detail |
| Enable reports cause `name-collision` | A command name collides with a built-in trigger, or a tool/command is declared twice by the same plugin | Rename the contribution |
| Tool call fails with a serialisation error | Handler returned a value `JSON.stringify` cannot serialise (e.g. cyclic) | Return a plain object or string |
| `[err] Plugin subsystem is disabled` | `plugins.enabled` is `false` | Set `plugins.enabled: true` in `ragent.json` |
| A capability is missing from the `ragent` object | The plugin's `plugins.permissions` grant omits that group | Add the capability (or `"*"`) to the grant list |

| Symptom | Cause | Fix |
|---|---|---|
| `mcp server transports` under Unsupported | Plugin declares an `mcpServers` entry shape the bridge cannot satisfy (e.g. an array of transports) | Often none - an inline object or a `"mcpServers": "./mcp.json"` file reference is bridged and connected as `<plugin-id>.<server>`; only the unparseable shape is reported |

---

## 15. Reference

### 15.1 The seven subcommands

```
list, add, remove, enable, disable, test, help
```

### 15.2 Host API v1 surface (quick list)

```
ragent.api_version
ragent.plugin_id
ragent.register_tool(def)
ragent.register_command(def)
ragent.config.get(key)
ragent.message.info(text) / .warn(text) / .error(text)
ragent.log(level, text)
ragent.plugin.read_text_file(relative_path)
```

### 15.3 Config keys

```
plugins.enabled
plugins.max_execution_ms
plugins.max_entry_ms
plugins.max_memory_mb
plugins.store_dir
plugins.permissions.<pluginid>
```

### 15.4 Fixture plugins (acceptance examples)

Eight fixture plugins under `assets/plugins/fixtures/` exercise the acceptance criteria and can be used as worked examples:

| Fixture | Dialect | Demonstrates |
|---|---|---|
| `codex-weather` | Codex | Manifest-declared tool + global-function handler |

| Fixture | Dialect | Demonstrates |
|---|---|---|
| `claude-todo` | Claude | Manifest tool + command with runtime `register_tool`/`register_command` handlers |
| `future-api` | Codex | `api_version: 99` version-mismatch refusal |
| `escape-attempt` | Codex | Filesystem-escape refusal (`read_text_file("../../outside.txt")`) |
| `loop-forever` | Codex | Entry-budget containment of `while (true) {}` |
| `claude-mcp` | Claude | `mcp_servers` recorded as unsupported |
| `collider` | Codex | Duplicate tool-name declaration (name collision) |
| `bad-manifest` | Codex | Truncated JSON manifest-parse failure |

### 15.5 Related commands and documents

- `/reload` - reload customisations after plugin files change.
- `/tools` - toggle tool visibility (plugin tools are always advertised while enabled).
- `ragent plugins <sub>` - CLI parity for the same subcommands.
- `docs/howtos/slashcommands/plugins.md` - the `/plugins` command reference.
- `docs/howtos/config.md` section 7.37 - the `plugins` config block.
- `specs/plugins/SPEC.md` - the full plugin-system specification.